

Technical Document #411: Software Engineering - Comprehensive Overview

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Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- API design defines how software components communicate.
- Refactoring improves code structure without changing behavior.
- Continuous deployment (CD) automatically deploys code changes to production.